

SWINBURNE UNIVERSITY OF TECHNOLOGY

DEPLOYMENT PORTFOLIO TASK 1

Pass to Distinction Implementation and Evidence Report

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Unit: SWE40006 Software Deployment and Evolution

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Declared target: Task 1.3 - Distinction

Source repository: https://github.com/RealSid08/SWE40006_Task1

Submission statement: Tasks 1.1, 1.2, and 1.3 were completed on Windows. The work covers application and WiX implementation, both MSI builds, and the installation, execution, and uninstallation checks required for Task 1.1. Both final packages were then reinstalled so the applications remain available on the PC. The Distinction implementation explicitly packages three external runtime DLLs: Newtonsoft.Json, CsvHelper, and Humanizer.

1. Executive summary

This portfolio documents an end-to-end Windows deployment workflow using WiX Toolset 4.0.6 and .NET 10. Task 1.1 packages a Hello World console application; Task 1.2 introduces HashGuard Desktop, a WinForms checksum utility; and Task 1.3 extends the HashGuard installer with multiple external DLL dependencies. Both final MSI packages are installed, and their deployed application files are present.

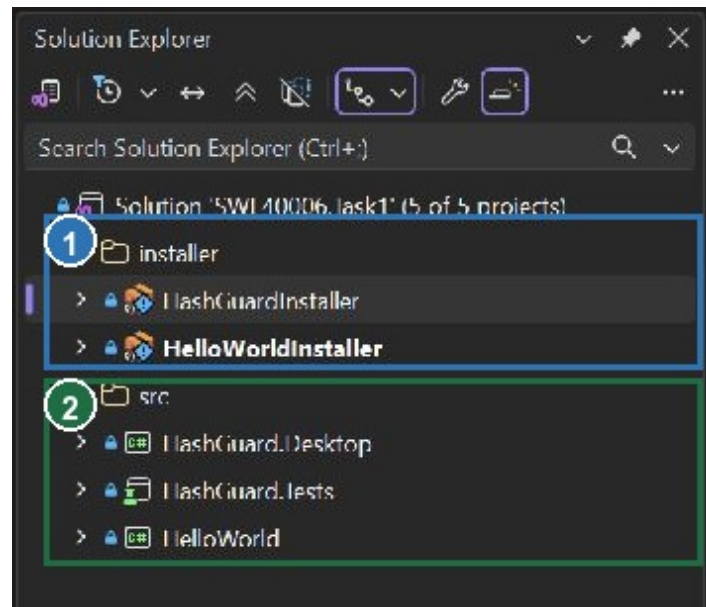
Outcome: Final automated verification produced 4 passing tests. Both MSI projects rebuilt with 0 warnings and 0 errors. The final Hello World and HashGuard installations completed successfully, their installed files were confirmed, and HashGuard is available from its Start Menu shortcut.

1.1 Setup environment and toolchain

Host environment		Native Windows 11 Home, 64-bit, build 26200
VM details		Not applicable - the assignment was completed directly on a Windows PC rather than a macOS virtual machine
Solution		SWE40006.Task1.sln containing five application, test, and installer projects
Build target		Release configuration; win-x64 application publish and x64 MSI packages
Component	Installed version	Role
Visual Studio	Community 2026 18.9.0	.NET desktop workload and solution/project authoring
.NET	SDK 10.0.400; Desktop Runtime 10.0.11	Build, test, publish, and execute the applications
WiX Toolset	CLI and SDK 4.0.6	Compile the two x64 MSI packages
HeatWave	Visual Studio extension installed	WiX project templates and Visual Studio integration

1.2 Visual Studio solution setup

The complete SWE40006.Task1 solution was opened in Visual Studio and verified to contain both WiX installer projects, along with the Hello World, HashGuard Desktop, and HashGuard test projects. Keeping the application, tests, and deployment source in one solution supports independent builds while maintaining a clear project structure.



- 1 Installer solution folder containing HashGuardInstaller and HelloWorldInstaller.
- 2 Source solution folder containing HashGuard.Desktop, HashGuard.Tests, and HelloWorld.

Figure 1. Annotated Visual Studio Solution Explorer showing the two WiX installer projects and the three application/test projects.

1.3 Step-by-step setup and build workflow

1. **Set up the Windows toolchain:** The environment used Visual Studio Community 2026 18.9.0 with the .NET desktop development tools, .NET SDK 10.0.400, .NET Desktop Runtime 10.0.11, WiX Toolset 4.0.6, and the HeatWave Visual Studio extension.
2. **Open and verify the solution:** SWE40006.Task1.sln was opened in Visual Studio, where HelloWorld, HashGuard.Desktop, HashGuard.Tests, HelloWorldInstaller, and HashGuardInstaller were confirmed as loaded.
3. **Restore, build, and test the applications:** Pinned NuGet packages were restored, the C# projects were built in Release mode, and the HashGuard test project was run. The final test result was four passes with no failures or skipped tests.
4. **Publish the deployable application files:** Hello World and HashGuard were published for win-x64 as framework-dependent applications. This produced a stable Release payload for each installer containing the executable, application DLL, dependency manifest, and runtime configuration.

5. **Author the WiX packages:** The .wxs files define the installation directories, application files, product identity, upgrade behaviour, and shortcuts. Task 1.3 adds separate WiX components for Newtonsoft.Json.dll, CsvHelper.dll, and Humanizer.dll.
6. **Build both MSI packages:** HelloWorldInstaller and HashGuardInstaller were compiled in Release mode. Both final builds produced the expected x64 MSI file with 0 warnings and 0 errors.
7. **Install and inspect the deployed files:** Each MSI was installed with Windows Installer verbose logging, followed by exit-status and installed-directory checks. Both installation logs ended with status 0, and the expected files were present.
8. **Execute and verify the installed applications:** Hello World was run from its installed directory and its banner was captured. The installed HashGuard application was launched to calculate both checksums for the sample payload and confirm that a history record was saved. The Task 1.1 uninstall check was also completed before the final packages were reinstalled.

1.4 Console proof of successful builds and installs

Final Release build output - evidence/31 and evidence/32

```
HelloWorldInstaller -> ...\\SWE40006-HelloWorld-1.0.0-x64.msi
Build succeeded.
    0 Warning(s)
    0 Error(s)

HashGuardInstaller -> ...\\HashGuard-Desktop-1.0.0-x64.msi
Build succeeded.
    0 Warning(s)
    0 Error(s)
```

Final Windows Installer log excerpts - evidence/43 and evidence/44

```
Windows Installer installed the product. Product Name: SWE40006 Hello World.
Installation success or error status: 0.
MainEngineThread is returning 0

Windows Installer installed the product. Product Name: HashGuard Desktop.
Installation success or error status: 0.
MainEngineThread is returning 0
```

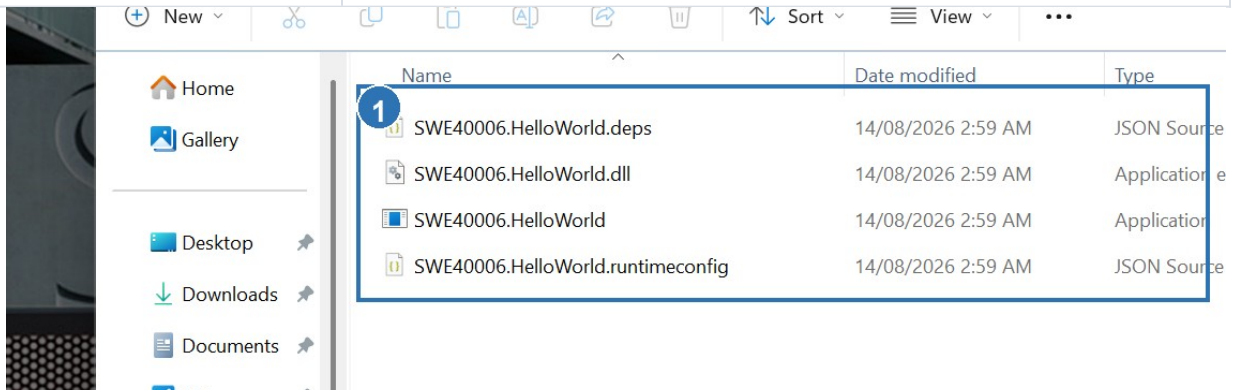
1.5 Reproducibility controls

WiX SDK and extension versions are pinned in the .wixproj files, and the NuGet dependency versions are locked. Stable product and upgrade GUIDs provide consistent package identity. Final test, build, installation, execution, and metadata logs are retained in the evidence directory, while both MSI SHA-256 hashes are recorded in evidence/33-msi-metadata.txt.

2. Task 1.1 - Pass: Hello World MSI

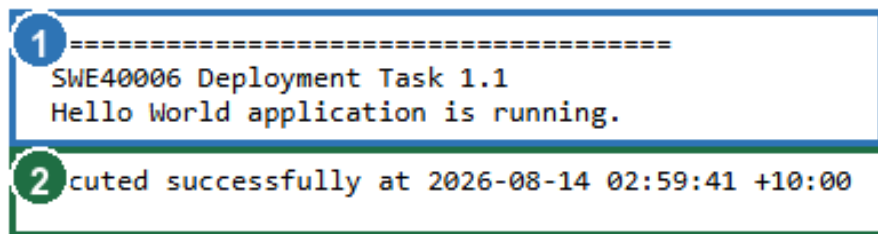
The Pass task uses a small .NET console program that prints an assignment-specific banner. A WiX x64 package installs the published executable and its three companion runtime files to %LOCALAPPDATA%\Programs\SWE40006 Hello World. The MSI was installed silently, the deployed executable was run from that directory, and the output was captured. The required uninstall check was completed before the final MSI was reinstalled, leaving Hello World available on the PC.

MSI	SWE40006-HelloWorld-1.0.0-x64.msi
Product version	1.0.0
Product code	{40C76FDB-606D-4DE6-9B17-F06149341B0A}
Install result	Exit code 0; four application files present
Execution result	Installed application printed the expected Task 1.1 banner
Final state	Package reinstalled; four application files remain present



- 1 The installed directory contains the Hello World executable, application DLL, dependency manifest, and runtime configuration.

Figure 2. Annotated installed-directory evidence for the Task 1.1 Hello World package.



- 1 The installed executable printed the assignment-specific Task 1.1 banner.
- 2 The captured process completed successfully with exit code 0.

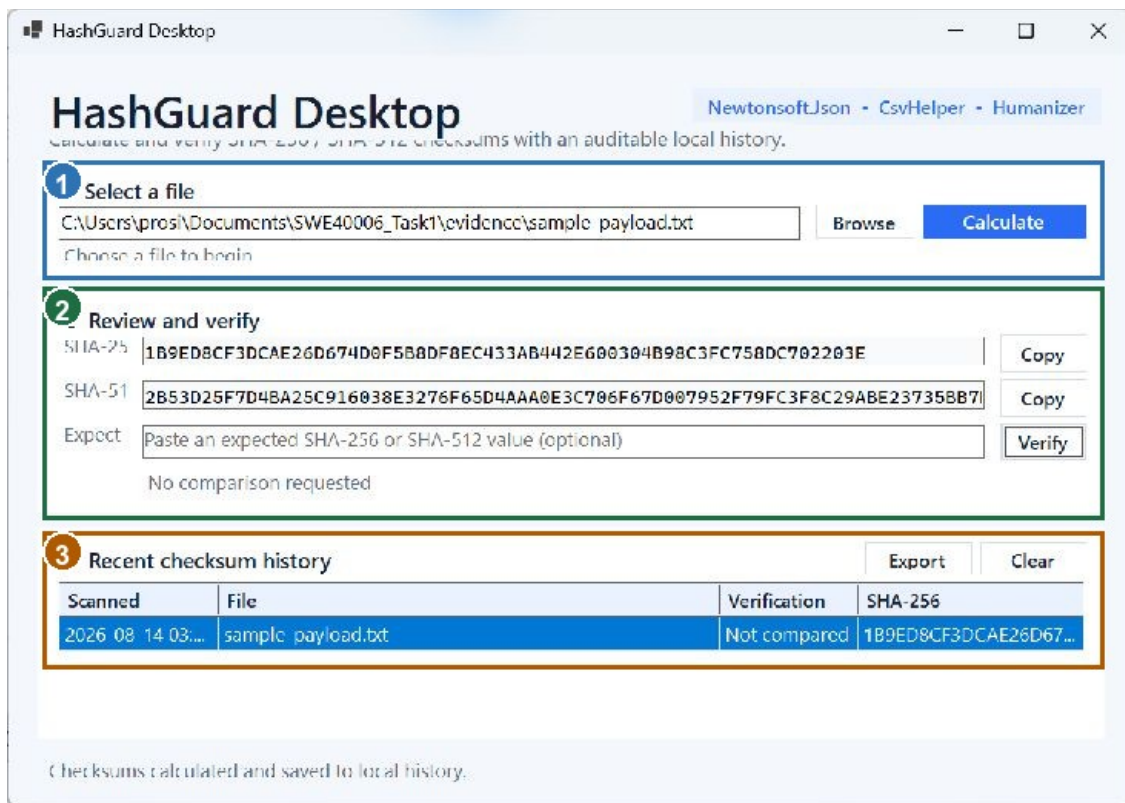
Figure 3. Annotated console output from the installed Hello World executable.

3. Task 1.2 - Credit: HashGuard Desktop

The Credit task introduces HashGuard Desktop, an original C# WinForms application for calculating SHA-256 and SHA-512 file checksums. Its implementation includes asynchronous hashing, optional comparison with an expected hash, a local JSON audit history, and CSV export. The interface also identifies the external dependencies and displays complete hash values for copying and verification.

3.1 Application design

Layer	Implementation	Responsibility
Presentation	MainForm.cs	File selection, progress, hash display, comparison, history, and export actions
Hashing	HashService.cs	Buffered asynchronous SHA-256/SHA-512 computation using IncrementalHash
Persistence	HistoryStore.cs	Auditable local JSON history using Newtonsoft.Json
Export	CsvExportService.cs	CSV output through CsvHelper
Presentation helper	Humanizer	Readable dates and display text



- 1 Input file: evidence/sample-payload.txt.
- 2 HashGuard calculated and displayed complete SHA-256 and SHA-512 values.
- 3 The application stored the completed calculation in its local audit history.

Figure 4. Annotated execution evidence from the installed HashGuard application.

3.2 Core checksum implementation

C# excerpt - HashService.ComputeAsync

```
using var sha256 = IncrementalHash.CreateHash(HashAlgorithmName.SHA256);
using var sha512 = IncrementalHash.CreateHash(HashAlgorithmName.SHA512);
await using var stream = new FileStream(
    filePath,
    FileMode.Open,
    FileAccess.Read,
    FileShare.Read,
    BufferSize,
    FileOptions.Asynchronous | FileOptions.SequentialScan);

var buffer = new byte[BufferSize];
long totalRead = 0;
int bytesRead;

while ((bytesRead = await stream.ReadAsync(buffer, cancellationTokens)) > 0)
{
    sha256.AppendData(buffer, 0, bytesRead);
    sha512.AppendData(buffer, 0, bytesRead);
    totalRead += bytesRead;
    progress?.Report(fileInfo.Length == 0 ? 100 : (int)(totalRead * 100 / fileInfo.Length));
}

progress?.Report(100);
return new HashResult(
    fileInfo.FullName,
    fileInfo.Length,
    Convert.ToHexString(sha256.GetHashAndReset()),
    Convert.ToHexString(sha512.GetHashAndReset()));
```

HashService streams each file in 1 MiB buffers and appends every buffer to both hash algorithms. This avoids loading the entire file into memory and allows the interface to display progress for large inputs.

3.3 Automated verification

```
HashGuard.Desktop -> C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Desktop\bin\Release\net10.0-windows\HashGuard.Desktop.dll
HashGuard.Tests -> C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Tests\bin\Release\net10.0-windows\HashGuard.Tests.dll
Test run for C:\Users\prosi\Documents\SWE40006_Task1\src\HashGuard.Tests\bin\Release\net10.0-windows\HashGuard.Tests.dll (.NETCoreApp,Version=v10.0)
A total of 1 test files matched the specified pattern.
1 .d! - Failed: 0, Passed: 4, Skipped: 0, Total: 4, Duration: 68 ms - HashGuard.Tests.dll (net10.0)
```

1 The final Release test run completed with 4 passed, 0 failed, and 0 skipped tests.

Figure 5. Annotated console output from the final automated test run.

Test	Expected behaviour	Result
Known hashes	SHA-256 and SHA-512 match known values	Passed
Mismatch verification	Malformed or different expected hash is rejected	Passed
JSON round trip	Newtonsoft.Json persists and restores a history record	Passed
CSV export	CsvHelper writes an export containing the history data	Passed

4. Task 1.3 - Distinction: external dependencies

Distinction criterion: The Distinction criterion is addressed by authoring multiple runtime dependencies as individual WiX components. The final HashGuard installation was verified to contain Newtonsoft.Json.dll, CsvHelper.dll, and Humanizer.dll alongside the application.

4.1 Runtime dependency set

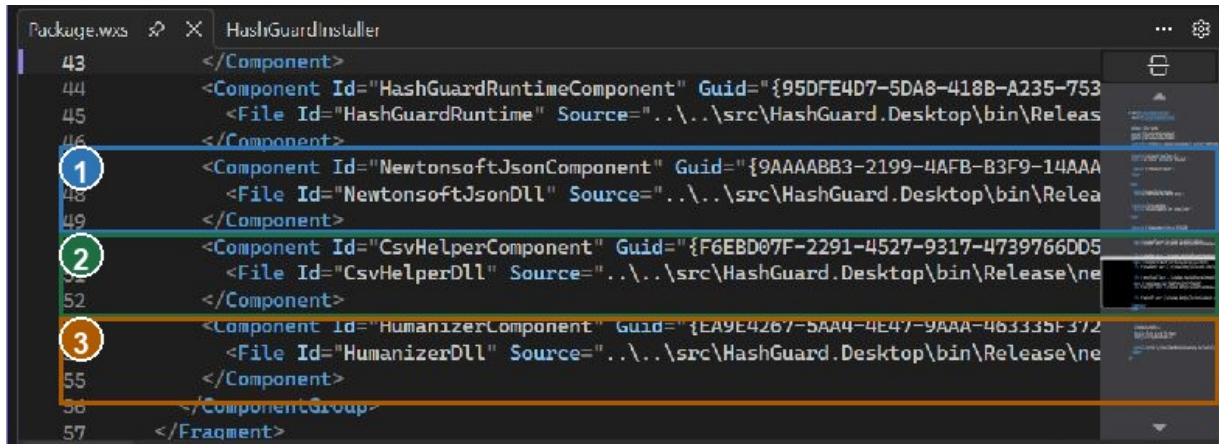
Dependency	Pinned version	Use in HashGuard
Newtonsoft.Json	13.0.4	Serializes and deserializes checksum history
CsvHelper	33.1.0	Exports the audit history to CSV
Humanizer.Core	3.0.10	Formats readable time and display values

4.2 WiX authoring

WiX v4 excerpt - explicit external DLL components

```
<Component Id="NewtonsoftJsonComponent" Guid="{9AAAABB3-2199-4AFB-B3F9-14AAAB88A66A}">
  <File Id="NewtonsoftJsonDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\Newtonsoft.Json.dll" KeyPath="yes" />
  <Component Id="CsvHelperComponent" Guid="{F6EBD07F-2291-4527-9317-4739766DD50B}">
    <File Id="CsvHelperDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\CsvHelper.dll" KeyPath="yes" />
    <Component Id="HumanizerComponent" Guid="{EA9E4267-5AA4-4E47-9AAA-463335F372FD}">
      <File Id="HumanizerDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\Humanizer.dll" KeyPath="yes" />
    
```

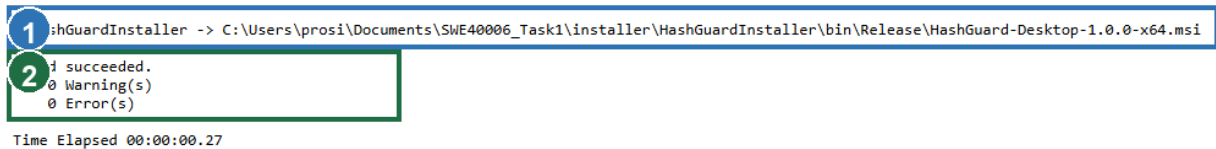
Each external file has its own stable component GUID and KeyPath. The same package also defines the application executable, application DLL, dependency manifest, runtime configuration, installation directory, and Start Menu shortcut.



- 1 Package.wxs defines Newtonsoft.Json.dll as an explicit WiX file component.
- 2 Package.wxs defines CsvHelper.dll as an explicit WiX file component.
- 3 Package.wxs defines Humanizer.dll as an explicit WiX file component.

Figure 6. Annotated Visual Studio view of the Package.wxs external-DLL components.

4.3 Build and installed-file evidence



- 1 The Release build produced HashGuard-Desktop-1.0.0-x64.msi.
- 2 The WiX build succeeded with 0 warnings and 0 errors.

Figure 7. Annotated console output from the final HashGuard MSI build.

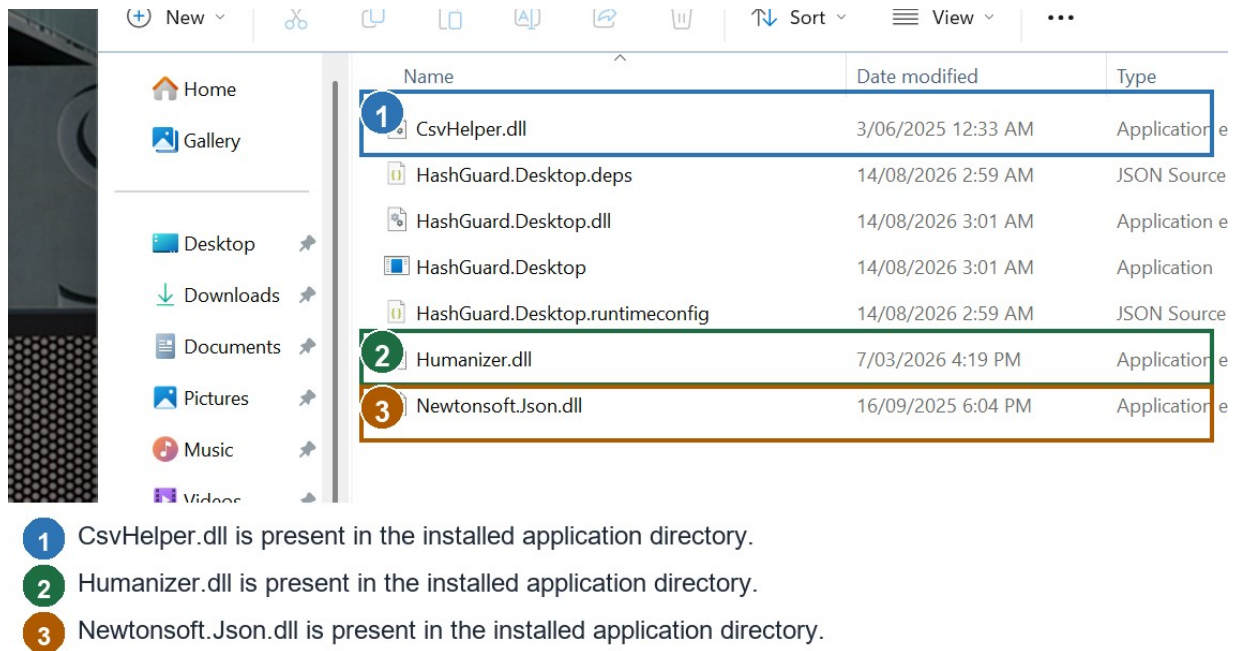


Figure 8. Annotated installed-directory evidence for the three external runtime DLLs.

4.4 MSI identity and installation

MSI	HashGuard-Desktop-1.0.0-x64.msi
Product code	{C57B16A8-E9E5-4EE9-8020-EE069EB409A7}
Upgrade code	{23AB1A31-A44A-41FE-BBDA-5245030BAD76}
Size	1,093,632 bytes
SHA-256	8B44ED575B560DD0C996AEB5CC1CF06C06DE8C9A3F31D669A3C50134AB488ED1
Final install	Exit code 0; 7 files plus Start Menu shortcut
Execution	Installed application launched and processed evidence/sample-payload.txt

5. Verification and evidence matrix

Requirement	Evidence	Verified result
Visual Studio setup	47-visual-studio-solution-and-wix.png	Five projects loaded in Visual Studio; Package.wxs opened for inspection
Toolchain	27-toolchain.txt, 28-dotnet-tools.txt, 29-visual-studio-workload.json	.NET, WiX, HeatWave/Visual Studio workload recorded
Pass build	31-final-hello-msi-build.log	0 warnings; 0 errors
Pass lifecycle	06-hello-install.log, 08-hello-execution.log, 41-hello-uninstall.log, 43-restored-hello-install.log	Required installation, execution, and uninstallation verified; final package reinstalled
Credit tests	30-final-tests.log	4 passed; 0 failed
Distinction build	32-final-hashguard-msi-build.log	0 warnings; 0 errors
Distinction install	44-restored-hashguard-install.log, 20-hashguard-final-files.txt	Exit code 0; 3 external DLLs present in final installation
MSI metadata	33-msi-metadata.txt	Product identity and SHA-256 hashes recorded

5.1 Manual checksum result

Input	evidence/sample-payload.txt
SHA-256	1B9ED8CF3DCAE26D674D0F5B8DF8EC433AB442E600304B98C3FC758DC702203E
SHA-512	2B53D25F7D4BA25C916038E3276F65D4AAA0E3C706F67D007952F79FC3F8C29AB E23735BB7D6E1A41C3AE4A89D9EE1694A82E1894C4A9EBC3F1B121A640C47AF
History	One record written to the local JSON audit history

6. Troubleshooting and engineering decisions

6.1 WinForms designer serialization warning

Initial compilation produced WFO1000 because WinForms designer serialization interpreted public form state as design-time content. The form was refactored so its application services and state are private readonly fields initialized in the constructor. This separated runtime state from designer serialization and restored a clean build.

6.2 WiX v4 UI migration

The initial WiX build used the WiX v3 UIRef pattern and failed with WIX0094 because that identifier is protected in WiX v4. The package was migrated to the WiX v4 UI extension namespace using ui:WixUI with WixUI_InstallDir and INSTALLFOLDER. WixToolset.UI.wixext is pinned to version 4.0.6.

6.3 Installed-app runtime failure

The first installed HashGuard build terminated before displaying its window. The Windows Application Event Log identified a .NET 10 NotSupportedException caused by a transparent button border setting. Setting BorderSize to 0 corrected the custom styling, after which the tests, publish, MSI build, installation, and installed-application execution all completed successfully.

6.4 MSI validation policy

The first non-elevated WiX build could not run ICE validation under the available Windows policy. Verification therefore used Windows Installer COM metadata, SHA-256 hashes, real silent installations, installed-file checks, execution from each deployed directory, and MSI log review. The final reproducibility builds completed with 0 warnings and 0 errors.

6.5 Packaging decisions

Decision	Reason	Effect
Per-user MSI	Avoids unnecessary elevation for a student utility	Installs under LocalAppData and supports silent testing
Explicit DLL components	Makes Distinction dependency evidence unambiguous	Each external DLL has a stable component GUID and key path
Framework-dependent publish	Keeps the MSI small while targeting the installed Desktop Runtime	Runtimeconfig and deps files are deployed with the application

7. Reproduction procedure

The following commands were run from the repository root on Windows with .NET 10 and WiX Toolset 4.0.6 installed. Pinned project dependencies allow the same commands to reproduce the test, publish, build, and installation results.

PowerShell - build and test

```
dotnet test src/HashGuard.Tests/HashGuard.Tests.csproj -c Release
dotnet publish src/HashGuard.Desktop/HashGuard.Desktop.csproj -c Release -r win-x64 --self-contained false
dotnet build installer/HelloWorldInstaller/HelloWorldInstaller.wixproj -c Release
dotnet build installer/HashGuardInstaller/HashGuardInstaller.wixproj -c Release
```

PowerShell - install

```
msiexec /i installer\HelloWorldInstaller\bin\Release\SWE40006-HelloWorld-1.0.0-x64.msi /qn /L*v evidence\43-restored-hello-install.log
msiexec /i installer\HashGuardInstaller\bin\Release\HashGuard-Desktop-1.0.0-x64.msi /qn /L*v evidence\44-restored-hashguard-install.log
```

7.1 Expected outputs

Step	Success condition	Final observed result
Test	All automated tests pass	4/4 passed
Build	Both MSI projects succeed	0 warnings; 0 errors
Install	MSI exits 0 and files appear	Both exits 0; 4 and 7 files respectively
Run	Installed executable starts	Hello banner and HashGuard checksum result captured

8. References

FireGiant. HeatWave Community Edition and Visual Studio integration. <https://docs.firegiant.com/heatwave/>

FireGiant. WiX Toolset v4 UI extension (WixUI). <https://docs.firegiant.com/wix/tools/wixext/wixui/>

Microsoft. IncrementalHash class. <https://learn.microsoft.com/dotnet/api/system.security.cryptography.incrementalhash>

NuGet. Newtonsoft.Json 13.0.4. <https://www.nuget.org/packages/Newtonsoft.Json/13.0.4>

NuGet. CsvHelper 33.1.0. <https://www.nuget.org/packages/CsvHelper/33.1.0>

NuGet. Humanizer.Core 3.0.10. <https://www.nuget.org/packages/Humanizer.Core/3.0.10>